Regularization

Gov 51: Data Analysis and Politics

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Week 9
Starting March 24, 2026

Exams coming back; today we start prediction and ML

- Exams being graded — back soon
- Today: **prediction and machine learning**
- **Project Proposal (Milestone 1) due Thursday, March 26 by 11:59 pm**

Your Project:
Finding a Question, Finding Data

The project is where you use everything we've built

- Every problem set has been practice
- The project is the game
- Pick something you're genuinely curious about
- It doesn't need to be groundbreaking — it needs to be **careful, honest, and interesting to you**

Milestone 1 is a 1–2 page proposal: research question, study type, dataset, and brief plan

Every project falls into one of three buckets

Descriptive

What does the world look like?

Summary stats, visualizations, text analysis, mapping

Card et al. immigration speeches — that was a descriptive project

Predictive

Can we forecast an outcome?

Train/test splits, RMSE, LASSO — exactly what we learn this week

COMPAS recidivism prediction is this type

Causal

Does X cause Y?

Experiments, DiD, IV — we'll cover these in Weeks 11–12

Broockman & Kalla was a causal study

Your bucket determines your methods, your success metric, and how you interpret results

Good descriptive projects

- How has the racial wealth gap in the U.S. changed since 1990?
- What topics dominate congressional floor speeches about immigration?
- How do commute times vary across metropolitan areas?
- What share of defendants in Broward County are rearrested within two years?

Descriptive work is underrated. Good description is hard, and it's the foundation for everything else.

Good predictive projects

- Can we predict which defendants will be rearrested?
- Can we forecast which countries will experience civil war?
- Can we predict election outcomes from polling data?
- Can we identify fraudulent insurance claims from claim characteristics?

The key metric is **out-of-sample performance** — an overfit model that looks great on training data but fails on new data is worthless

Good causal projects

- Does increasing the minimum wage reduce employment?
- Does body camera adoption reduce police use of force?
- Does access to early childhood education improve long-term earnings?
- Did a specific policy change affect voter turnout?

You need a credible identification strategy — why is your comparison valid? We'll cover the methods in Weeks 11–12.

Good questions start with what bugs you

1. **Start with what bugs you**
   Housing costs frustrate you → "How have rents changed relative to wages?"
2. **Read the news with a social scientist's eye**
   "Crime is surging" — compared to what? That "compared to what?" is where questions live
3. **Narrow it down**
   "Inequality" is not a question. "Has the Gini coefficient changed differently in Medicaid-expansion vs. non-expansion states?" is.
4. **Start with the data**
   Find an interesting dataset, then ask what questions it can answer. No shame in that.

Eight sources cover most undergraduate projects

Source	Best for
IPUMS (ipums.org)	Census, ACS, demographics, income, housing
ANES (electionstudies.org)	Voting behavior, political attitudes
GSS (gss.norc.org)	Social attitudes since 1972
ICPSR / openICPSR	16,000+ datasets + AEA replication packages
Harvard Dataverse	Replication data from published papers
ProPublica	Criminal justice, investigations
FiveThirtyEight GitHub	Politics, sports (clean, small)
Opportunity Insights	**Income mobility by neighborhood (free CSVs)**

You can also collect your own: web scraping, text data, surveys, or LLM classification

Published studies post their data — use it, but ask a new question

- **Opportunity Insights** (opportunityinsights.org/data)
  Chetty et al.: tract-level mobility, earnings, incarceration, patents by race/income.
  Free CSV download, no registration.
- **Abramitzky & Boustan** (openICPSR project 120490)
  Immigrant mobility across two centuries — connects to Card et al. speeches we read
- **AEA replication packages** (openicpsr.org)
  Thousands of datasets from published economics papers — all free

Rule: if you use replication data, your question must be new — you're not replicating, you're investigating something the authors didn't

Load your data before you submit

- **Load your data into R before you submit**
  Make sure the file reads, the key variables exist, and there's enough variation
- **State your question in one sentence**
  If you can't, it's too big
- **Deciding between two questions? Submit both**
  That's what Milestone 1 is for — we'll give you feedback
- **It doesn't need to be ambitious**
  A well-executed descriptive analysis of something you care about is a great project

Due Wednesday, March 26 — 1–2 pages

Now: Prediction and Machine Learning

An algorithm could reduce crime by 25% — using math you already know

24.7%

reduction in crime with no change in jailing rates

Kleinberg, Lakkaraju, Lestre, Ludwig & Mullainathan (2018, *QJE*)

Judges make predictions every day — and the math can help

At every bail hearing, the judge predicts:

- Will this defendant show up for court?
- Will they commit a new crime before trial?

Kleinberg et al. results:

- Reduce crime by 24.7% at same jailing rate
- **Or** reduce jailing by 42% at same crime rate
- **Tools:** OLS, LASSO, Ridge — what we learn this week

Two experts, same data, opposite conclusions

Halliburton Co. v. Erica P. John Fund — \$5 billion securities case.

	Defense Expert	**Plaintiff Expert**
Peer selection method	S&P Energy + custom index	Analyst-report peers
Excess return	-2.9%	-3.7%
p-value	0.20	0.02
Conclusion	Not significant	Significant

Key problem: different peer choices $\rightarrow$ opposite answers

How do we build models where the data picks the variables, not the analyst?

The Bias-Variance Tradeoff

Adding predictors improves fit but worsens prediction

Before spring break:

- More predictors $\rightarrow$ better in-sample fit
- More predictors $\rightarrow$ worse out-of-sample predictions
- R^2 always goes up when you add variables — even useless ones

How do we build models that predict *new* data well, not just the data we already have?

In Week 6 we split the data once — today we generalize

Week 6: Single split

- 80% train, 20% test
- Fit on train, evaluate on test
- One RMSE estimate

Problem: result depends on which 20% you held out

Today: Cross-validation

- Split into k folds
- Each fold takes a turn as the test set
- Average k RMSE estimates

More stable estimate of out-of-sample performance

Same idea as Week 6 — but repeated k times for a more reliable answer

What makes a prediction "good"?

MSE measures total prediction error

For a single prediction:

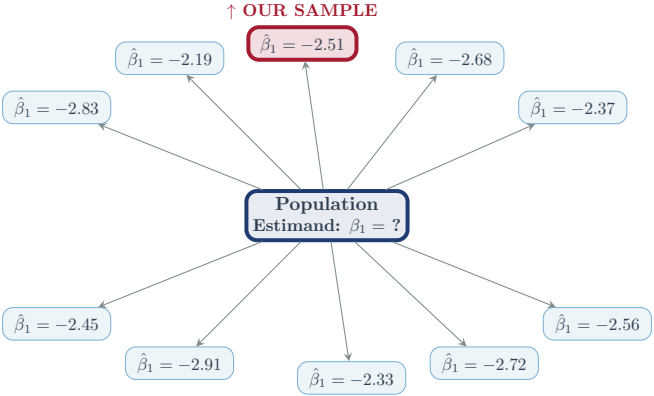
$$\text{Squared Error}_i = (\hat{Y}_i - Y_i)^2$$

Average over all predictions:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(\hat{Y}_i - Y_i)^2$$

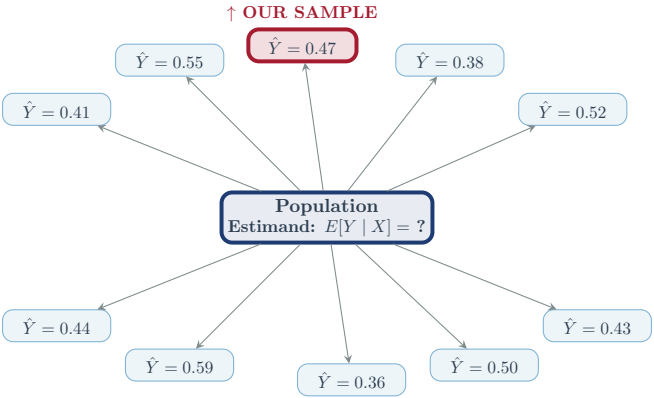
RMSE $= \sqrt{\text{MSE}}$ (back in original units)

Remember this? $\hat{\beta}_1$ is an estimator, β_1 is the estimand



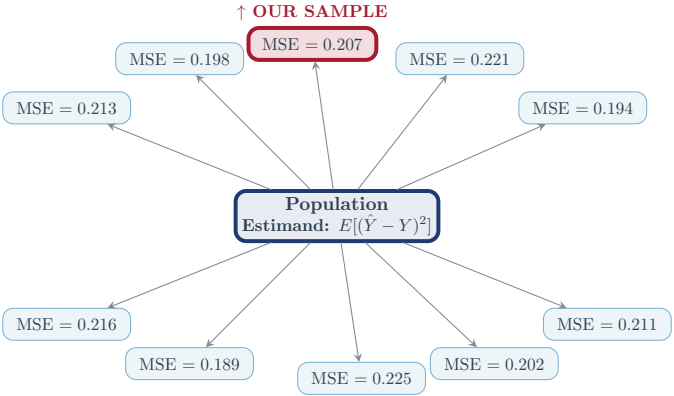
Each spoke is an **estimator**. The hub is the **estimand**.

Each sample also gives a different prediction $\hat{Y}$



Variance = how spread out the $\hat{Y}$'s are across samples

Our sample MSE is an estimator of the population MSE



Same logic as $\bar{X}$ estimating $E[X]$. The population MSE is what we decompose.

Bias means consistently wrong

Definition:

$$\text{Bias}(\hat{Y}) = E[\hat{Y}] - Y_{\text{true}}$$

- **Question:** over repeated samples, are predictions centered on the truth?
- **OLS:** unbiased — centered correctly on average
- **Key insight:** small bias can be useful if it buys stability

Variance means unpredictable

Definition:

$$\mathrm{Var}(\hat{Y}) = E\big[(\hat{Y} - E[\hat{Y}])^2\big]$$

- **Question:** over repeated samples, how much do predictions change?
- **Many parameters** $\rightarrow$ high variance
- **Few parameters** $\rightarrow$ low variance

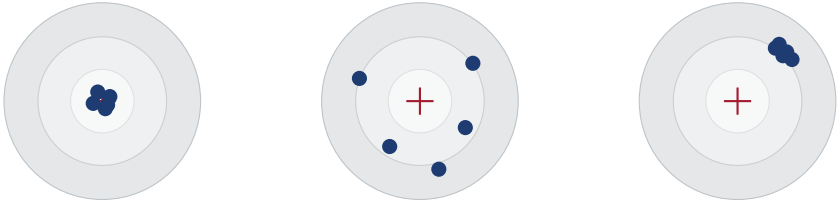
Noise is what no model can predict

Definition:

$$\epsilon_i = Y_i - Y_{\text{true},i}$$

- **Irreducible:** even the perfect model misses
- $E[\epsilon^2]$ = inherent randomness in outcomes
- No data or methods can reduce this term

Bias means consistently wrong; variance means scattered



Low Bias, Low Variance
The goal

Low Bias, High Variance
OLS with many predictors

High Bias, Low Variance
Regularized regression

The three sources of error are independent, so they add

In English:

1. Your prediction misses because you aimed at the wrong spot (bias)
2. Your prediction misses because your aim wobbles (variance)
3. Your prediction misses because the world is noisy (irreducible)

Independence: noise $\perp$ model choice $\Rightarrow$ the three terms add

Total miss = systematic miss + wobble + noise

Every prediction error comes from bias, variance, or noise

$$E\left[(\hat{Y} - Y)^2\right] = \underbrace{\text{Bias}^2}_{\text{systematic error}} + \underbrace{\text{Variance}}_{\text{instability}} + \underbrace{E[\epsilon^2]}_{\text{irreducible noise}}$$

We control bias and variance. We cannot control $E[\epsilon^2]$.

Where does the decomposition come from?

The "add and subtract the mean" trick:

$$E\big[(\hat{Y}-Y)^2\big] = E\big[\big(\underbrace{\hat{Y}-E[\hat{Y}]}_{\text{estimation error}} + \underbrace{E[\hat{Y}]-Y_{\text{true}}}_{\text{bias}} + \underbrace{Y_{\text{true}}-Y}_{\epsilon}\big)^2\big]$$

$$= E\big[(\hat{Y}-E[\hat{Y}])^2\big] + \big(E[\hat{Y}]-Y_{\text{true}}\big)^2 + E[\epsilon^2]$$

$$= \text{Variance} + \text{Bias}^2 + E[\epsilon^2]$$

Key step: cross terms vanish because $\epsilon \perp \hat{Y}$ and $E[\hat{Y}-E[\hat{Y}]] = 0$.

Gauss-Markov says OLS is the best *unbiased* estimator

Recall from Week 6:

OLS is **B**est **L**inear **U**nbiased **E**stimator (BLUE).

- "Best" = lowest variance *among unbiased estimators*
- **Unbiased $\neq$ best overall**

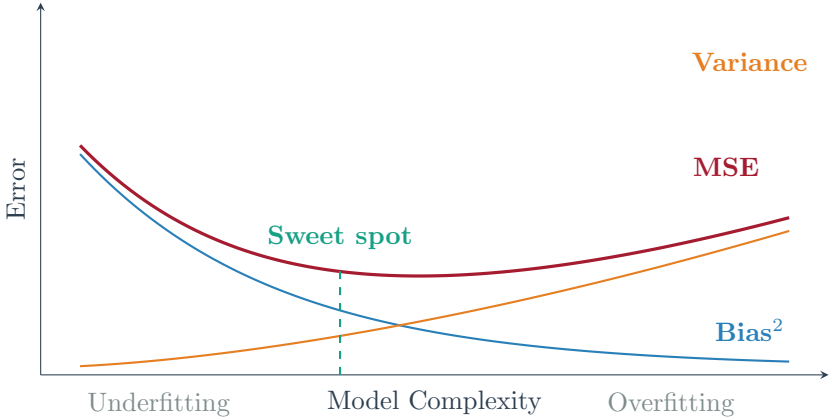
A little bias can buy a lot of stability

$$\text{MSE} \quad = \quad \text{Bias}^2 \; + \; \text{Variance}$$

Allow a *small* increase in $\text{Bias}^2 \rightarrow$
get a *large* decrease in Variance

Net effect: lower MSE, better predictions

The best model lives between underfitting and overfitting



The Big Idea: Penalize Complexity

What if we forced our regression coefficients to be smaller?

OLS has no constraints on coefficient size

Recall: OLS chooses $\hat{\beta}$ to minimize

$$\sum_{i=1}^{n} \left(Y_i - \hat{\alpha} - \sum_{j=1}^{p} \hat{\beta}_j X_{ij} \right)^2$$

- No limit on how large $\hat{\beta}_j$ can be
- With many predictors, OLS chases noise

Penalized regression forces coefficients to earn their place

$$\min_{\beta} \underbrace{\sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2}_{\text{fit the data}} + \underbrace{\lambda \cdot \text{Penalty}(\beta)}_{\text{keep coefficients small}}$$

- $\lambda = 0$: no penalty $\rightarrow$ OLS
- $\lambda \rightarrow \infty$: all coefficients shrink toward zero
- Cross-validation picks the right λ

Ridge Regression (1970)

Hoerl & Kennard invented Ridge at DuPont in 1970

- **Problem:** correlated predictors $\rightarrow$ wildly unstable OLS estimates
- **Solution:** add a squared penalty to shrink coefficients
- **Objection:** "Ridge is biased! Gauss-Markov says OLS is best!"
- **Response:** BLUE $\neq$ lowest MSE

Ridge shrinks all coefficients but never eliminates any

$$\min_{\beta} \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

- Penalty on the **sum of squared coefficients**
- Every variable stays in the model, just with smaller weight
- Never sets any coefficient *exactly* to zero

LASSO (1996)

Tibshirani invented LASSO at Toronto in 1996

Least **A**bsolute **S**hrinkage and **S**election **O**perator

- **Author:** Robert Tibshirani, Toronto $\rightarrow$ Stanford
- **Key move:** swap β_j^2 penalty for $|\beta_j|$ penalty
- **Result:** some coefficients shrink to **exactly zero**
- **LASSO selects**, not just shrinks

LASSO shrinks coefficients and sets some exactly to zero

$$\min_{\beta} \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

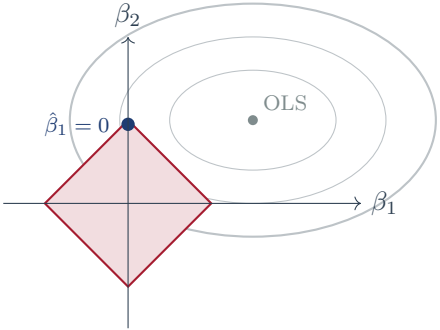
- Penalty on the **sum of absolute values**
- As λ increases, coefficients shrink
- Some hit exactly zero and drop out
- You get a sparse model: few variables, each one meaningful

LASSO tells you which variables the data actually needs

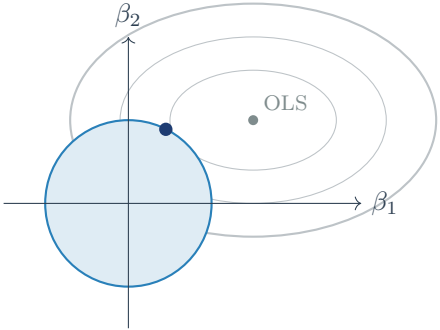
- **Prediction:** fewer noisy variables → less overfitting
- **Interpretation:** a short list of what matters
- **Honesty:** the data picks the model, not the analyst

Baker's principle: penalty selects variables → expert discretion removed

The geometry explains why LASSO zeros out coefficients



LASSO (L1)

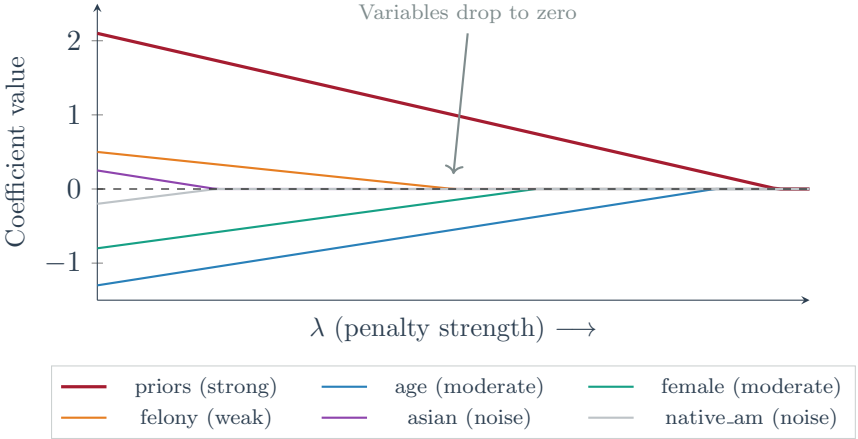


Ridge (L2)

Diamond corners → solutions land on axes → coefficients hit zero.

Circle has no corners → solutions never land exactly on an axis.

Prior record survives longest; noise variables drop out first



Elastic Net and the Full Picture

Elastic Net blends the LASSO and Ridge penalties

Zou & Hastie (2005)

$$\min_{\beta} \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 + \lambda \left[\alpha \sum_{j=1}^{p} |\beta_j| + (1 - \alpha) \sum_{j=1}^{p} \beta_j^2 \right]$$

- $\alpha = 1$: pure LASSO
- $\alpha = 0$: pure Ridge
- $0 < \alpha < 1$: blend of both

Ridge, LASSO, and Elastic Net differ in how they shrink

	Ridge	**LASSO**	**Elastic Net**
Penalty	$\sum \beta_j^2$	$\sum \lvert\beta_j\rvert$	Both
Shrinks coefficients?	Yes	Yes	Yes
Sets coefficients to zero?	No	Yes	Yes
Variable selection?	No	Yes	Yes
Handles correlation?	Well	Poorly	Well

How Do We Choose λ?

λ controls the bias-variance tradeoff

- λ too small $\rightarrow$ not enough penalty $\rightarrow$ overfitting (high variance)
- λ too large $\rightarrow$ too much penalty $\rightarrow$ underfitting (high bias)

Cross-validation estimates out-of-sample error for each λ, then picks the one that generalizes best.

k-fold cross-validation: each observation gets a turn

Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	$\rightarrow \text{RMSE}_1$
Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	$\rightarrow \text{RMSE}_2$
Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	$\rightarrow \text{RMSE}_3$
		$\vdots$			
Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	$\rightarrow \text{RMSE}_5$

Red = predict this fold **Blue** = train on these folds

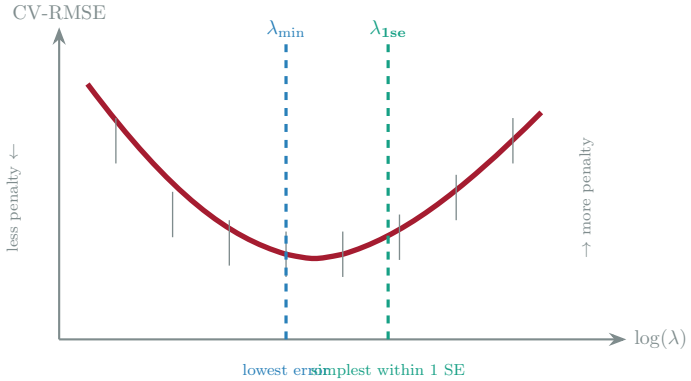
$$\text{CV-RMSE} = \frac{1}{5}(\text{RMSE}_1 + \cdots + \text{RMSE}_5)$$

Minimizing training error always picks $\lambda = 0$

- $\lambda = 0$ means no penalty $\rightarrow$ OLS
- OLS gives the best in-sample fit by definition
- But we already know OLS overfits!

Cross-validation estimates **out-of-sample** error $\Rightarrow$ optimal $\lambda > 0$

The CV error curve tells you which λ to use



- λ_{min}**:** minimizes CV error — best prediction
- $\lambda_{\textbf{1se}}$**:** simplest model within 1 SE of the minimum — more parsimonious

Two Cultures of Statistics

Breiman (2001): statistics has two cultures that barely talk

Data Modeling Culture	Algorithmic Culture
Specify a model *a priori*	Treat the process as unknown and complex
Estimate parameters (OLS)	Optimize prediction accuracy
Judge by R^2	Judge by out-of-sample error
Goal: understand the process	Goal: predict well
98% of statistics departments	Machine learning, industry

Before LASSO, the analyst picked the variables

- Researcher chooses variables based on theory and judgment
- Different researchers $\rightarrow$ different variable sets $\rightarrow$ different results
- Who's right? Whoever tells the most convincing story

After LASSO:

- Start with all candidate variables
- Let the penalty and cross-validation decide which ones stay

Back to the Courtroom: Baker (2025) and Expert Testimony

Expert witnesses exercise discretion over model specification

Baker (2025), *Berkeley Business Law Journal*

- Expert witnesses in litigation make predictions:
  - Stock price without the fraud (event studies)
  - Fair value of a company (bankruptcy)
  - Damages from antitrust violations
- Each side hires an expert who picks a different model

In Halliburton, different peer choices led to opposite conclusions

	Defense Expert	**Plaintiff Expert**
Peer selection	S&P Energy + custom index	Analyst-report peers
Excess return	-2.9%	-3.7%
p-value	0.20	0.02
Conclusion	Not significant	Significant

Stakes: class certification worth billions — hinged on peer selection

All three regularization methods converge on the same answer

- Baker applied LASSO, Ridge, and Elastic Net to 29 candidate peers
- LASSO kept 9, zeroed out 20

Method	**Excess Return**	p**-value**
LASSO	-3.2%	≈ 0.06
Ridge	-3.3%	≈ 0.06
Elastic Net	-3.2%	≈ 0.06

Result: data-driven answer falls between the two experts — all three methods agree

Penalized methods beat OLS when overfitting is the problem

	OLS	Ridge	LASSO	Elastic Net
Variables used	all 29	all 29 (shrunk)	9 of 29	12 of 29
Test RMSE	highest	lower	lowest	$\approx$ LASSO
Bias	zero	small	small	small
Variance	high	low	low	low

OLS has zero bias but high variance $\Rightarrow$ highest MSE
Penalized methods trade a little bias for a lot of stability

LASSO removes expert discretion: the data picks the variables

Baker's argument:

- Traditional expert testimony → subjective variable selection → dueling experts
- LASSO + cross-validation → data-driven selection → convergence

For us:

- Same penalized regression we learned today
- Applied to a $5+ billion securities case

A skeptic would say: biased estimators give the wrong answer

The objection:

- Gauss-Markov proves OLS is best
- Introducing bias means our coefficients are *systematically wrong*
- How can "wrong on purpose" be better?

The response:

- "Best" among *unbiased* estimators $\neq$ best overall
- MSE = Bias^2 + Variance
- A tiny bias that crushes variance *lowers* total error
- In the Halliburton case: biased methods *converged* while unbiased experts *disagreed*

How Uncertain Is Each Prediction?

The judge doesn't just want the best model — she wants to know how sure it is

Bail decision for defendant #4,872:

- **Point prediction:** 23% chance of rearrest
- **But how uncertain is "23%"?**
- Could easily be 10% or 40% — those imply different decisions

RMSE tells you how the model does *on average*
A prediction interval tells you how
uncertain *this specific prediction* is

Confidence intervals and prediction intervals answer different questions

Confidence interval

$\hat{Y} \pm t \times \mathrm{SE}_{\mathrm{mean}}$

- Where is the **average** Y for people like this?
- Shrinks with more data
- Estimation uncertainty only

Prediction interval

$\hat{Y} \pm t \times \mathrm{SE}_{\mathrm{pred}}$

- Where will **this person's** Y actually land?
- Cannot shrink below noise floor
- Estimation uncertainty + **noise**

$$\mathrm{SE}^2_{\mathrm{pred}} = \mathrm{SE}^2_{\mathrm{mean}} + \widehat{\sigma}^2 \quad \text{— the extra } \widehat{\sigma}^2$$

is the irreducible noise from the decomposition

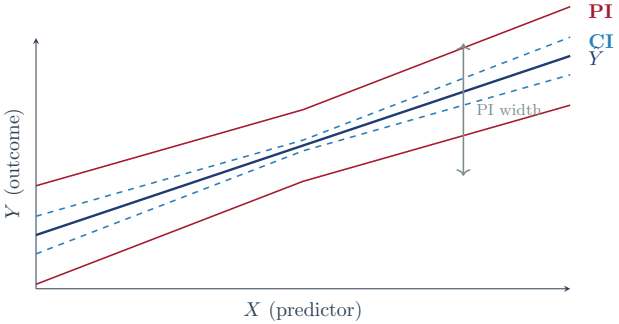
The PI formula has three terms — each connects to something you know

$$\hat{Y}_0 \;\pm\; t \times \sqrt{\widehat{\text{MSE}} \times \left(\underbrace{1}_{\text{noise}} + \underbrace{\frac{1}{n}}_{\text{intercept}} + \underbrace{\frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2}}_{\text{slope / leverage}} \right)}$$

Term	What it captures	Connection
1	New person's ε (iid draw)	Irreducible noise
$1/n$	Uncertainty about intercept	Shrinks with n — like $\text{SE}_{\bar{Y}}$
$(x_0 - \bar{x})^2 / \cdots$	Uncertainty about slope	Extrapolation penalty

Even with $n \to \infty$, the **1** never goes away — that's what "irreducible" means

The prediction interval is always wider than the confidence interval



- **CI** (dashed): narrows with more data — estimation uncertainty
- **PI** (solid): has a **floor** — irreducible noise never goes away

In R: one argument gives you prediction intervals

Confidence interval (for the mean):

```
predict(fit, newdata, interval = "confidence")
```

Prediction interval (for an individual):

```
predict(fit, newdata, interval = "prediction")
```

	fit	lwr	upr
CI for mean	\$1,117	\$1,095	\$1,139
PI for individual	\$1,117	\$542	\$1,692

Same point prediction. CI width: \$44. PI width: \$1,150.

For LASSO and Ridge, prediction intervals are an open problem

The problem:

- OLS has a clean PI formula
- LASSO/Ridge are biased by design
- Standard formula doesn't apply
- No `interval = "prediction"` in `glmnet`

What you can do:

- **RMSE:** average prediction error
- **Bootstrap:** resample, refit, predict $\rightarrow$ spread = uncertainty
- **Conformal prediction:** works for any model (Rambachan, MIT)

For your projects: OLS prediction intervals where possible, RMSE comparison across methods always

What's Next

Thursday: you build prediction models for recidivism

Problem Set 3: Prediction and Regularization

- COMPAS recidivism dataset (ProPublica)
- Fit OLS, then expand to ∼200 interaction terms
- Watch OLS overfit
- Fix it with Ridge, LASSO, and Elastic Net
- Cross-validation by hand, then with `cv.glmnet()`
- Discuss Baker and algorithmic fairness

Next week: same regression, different purpose

Week 10: From prediction to causal inference

- We've been asking: what will happen?
- Next: why did it happen?
- Covariates reduce MSE in prediction
- Covariates reduce *bias* in causal inference

The data should pick the model, not the analyst.

A little bias buys a lot of stability.